Pivot positions in rref: $\begin{bmatrix} \boxed{1} & 0 & -1 & 0 \\ 0 & \boxed{1} & 2 & 0 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$

Pivot positions in original form: $\begin{bmatrix} \boxed{1} & 3 & 5 & 7 \\ 3 & \boxed{5} & 7 & 9 \\ 5 & 7 & 9 & \boxed{1} \end{bmatrix}$

The pivot columns are: 1, 2 and 4

5) Describe the possible echelon forms of a nonzero 2×2 matrix. Use the symbols □, *, and 0, as in the first part of Example 1.

$\boxed{\text{Soln:}}$ For a matrix to be in echelon form, it must satisfy the following conditions:

1. All the nonzero rows must be above all zero rows.
2. The leading entry of each row must be in a column to the right of the columns of the leading entries of all rows above it.
3. For each row having a leading entry, the corresponding column has all 0's below it.

Let, row 1 have it's leading entry (□) in column 1. By condition 3, row 2 column 1 is 0. row 2 column 2 can have a leading entry (□). Condition 2 and 3 are satisfied in this case. Row 1, Column 2 can have any of the 2 values * or 0. The corresponding matrices are: $\begin{bmatrix} \square & * \\ 0 & \square \end{bmatrix}$, $\begin{bmatrix} \square & 0 \\ 0 & \square \end{bmatrix}$

row 2 can choose to not have a leading entry. In that case, the corresponding matrices are:

$\begin{bmatrix} \square & * \\ 0 & * \end{bmatrix}$, $\begin{bmatrix} \square & 0 \\ 0 & * \end{bmatrix}$, $\begin{bmatrix} \square & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} \square & * \\ 0 & 0 \end{bmatrix}$

row 1 has a leading entry (□) in column 2, that is also possible.
Then, by defn. of leading entry, row 1 column 1 is 0.